SUMIT YADAV

Graduate Software Engineer | Python | Django | PostgreSQL

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LinkedIn | GitHub | Portfolio

PROFILE

MSc Advanced Computer Science graduate with hands-on full-stack development experience across food technology and UK vehicle analytics. Built and deployed Django applications with Python, PostgreSQL, APIs and CI/CD. Processed 42.6 million MOT records for Carverse Check and wrote 376 automated tests for CulinaAI.

TECHNICAL SKILLS

Languages and web: Python, JavaScript, PHP, SQL, HTML, CSS, Django, Bootstrap, RESTful APIs

Databases and data: PostgreSQL, MySQL, Neon, DuckDB, dbt, data modelling, ETL pipelines, data validation

Cloud and delivery: AWS S3, AWS IAM, Render, Docker, GitHub Actions, CI/CD, Prefect, Git, GitHub, GitLab

Testing: Django TestCase, automated and regression testing, Playwright, axe-core, Lighthouse CI, dbt tests

Engineering: Object-oriented programming, system design, requirements engineering, authentication, RBAC, Sentry

SOFTWARE ENGINEERING PROJECTS

Carverse Check | UK Vehicle History and MOT Analytics **Sep 2026 - Oct 2026**

Python, Django, DuckDB, dbt, Prefect, PostgreSQL, Neon, DVSA API, Playwright, GitHub Actions

Live website | GitHub repository

- Processed **42.6 million MOT tests and 91.8 million defect records** by loading DVSA CSV files into DuckDB, deduplicating them with dbt and modelling data for vehicle comparisons.
- Published statistics for **17,460 vehicle groups** to Neon PostgreSQL using transactional updates, so the live Django site on Render only receives complete results.
- Checked outputs against DVSA figures and ran **49 dbt data tests** before publishing; used Prefect to retry failed pipeline runs and skip unchanged files.
- Built DVSA MOT History API reports with mileage trends, defect history and peer comparisons. Added rate limiting, configuration checks, network error handling and secret-safe logs, backed by **103 Django tests** and **17 Playwright tests** in CI.

CulinaAI | AI-Assisted Recipe Generation and Meal Personalisation **May 2026 - Sep 2026**

MSc dissertation and independent project | Python, Django, PostgreSQL, OpenAI API, AWS S3, GitHub Actions

Live website | GitHub repository

- Delivered **29 functional and 10 non-functional requirements** and passed **64/64 test scenarios** by building Django views, PostgreSQL models and recipe-personalisation workflows.
- Built **376 automated Django tests** and ran them through GitHub Actions CI/CD to check changes across recipe generation, account management and meal planning.
- Got **30/30 benchmark recipes through validation on the first attempt** by fixing allergy and equipment-check defects, refining AI regeneration feedback and adding regression tests.
- Improved food-data matching from **21/30 to 28/30 recipes** through ingredient aliases; processed **2,759 UK food records** with Python scripts to calculate nutrition and recipe costs in application code.
- Built a **10-check recipe validation engine** covering allergies, dietary needs, ingredients and cooking constraints; returned specific feedback for failed checks and a safe fallback when retries failed.
- Deployed on Render with AWS S3 storage, least-privilege IAM permissions, Sentry monitoring, email verification and daily AI usage limits.

CarVerse | Car Dealership Web Application **2023 - 2024**

BSc final-year project | PHP, MySQL, JavaScript, HTML, CSS | GitHub repository

- Built a **40+ page** PHP/MySQL dealership website with administrator CRUD, image uploads and role-based login; added vehicle search, comparison and booking workflows.

EDUCATION

MSc Advanced Computer Science, University of Leicester, UK Sep 2025 - Sep 2026

BSc Computer Science, PTVA's Mulund College of Commerce, India 2021 - 2024

CGPA: **8.53/10**